

MATH1141: Higher Mathematics 1B - Algebra

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Chapter 1

Vector Spaces

A field is a non-empty set $\mathbb{F}$ which has two binary operations $+$, $\cdot$, and satisfies all 12 number laws.

1.1 Examples of fields

- $\mathbb{Q}, \mathbb{R}, \mathbb{C}$
- $\mathbb{Q}(\sqrt{3}) := \{a + b\sqrt{3} \mid a, b \in \mathbb{Q}\}$
- $\mathbb{Q}(i) := \{a + bi \mid a, b \in \mathbb{Q}\}$, where $i = \sqrt{-1}$
- Every prime number p defines a field

$$\mathbb{F}_p = GF(p) := 0, 1, 2, \dots, p - 1$$

1.2 Definition of Vector Space

A vector space over a field F (the set of scalars) is a non-empty set V of elements, vectors, for which addition and scalar multiplication are defined and which obey the following 10 axioms.

1. **Closure under addition:** adding two vectors yields a vector.
2. **Associative law of addition:** $(\mathbf{u} + \mathbf{b}) + \mathbf{w} = \mathbf{u} + (\mathbf{b} + \mathbf{w})$.
3. **Commutative law of addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
4. **Existence of zero:** there is an element $\mathbf{0}$ in V called the zero vector that satisfies the property $\mathbf{v} + \mathbf{0} = \mathbf{v}$, $\forall \mathbf{v} \in V$.
5. **Existence of negative:** for each $\mathbf{v} \in V$ there exists an element $\mathbf{w} \in V$ such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$.
6. **Closure under scalar multiplication:** for all $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, $\lambda \mathbf{v} \in V$.
7. **Associative law of scalar multiplication:** for all $\lambda, \mu \in \mathbb{F}$ and $\mathbf{v} \in V$,

$$\lambda(\mu \mathbf{v}) = (\lambda\mu)\mathbf{v}.$$

8. For all $\mathbf{v} \in V$, $1\mathbf{v} = \mathbf{v}$.

9. **Scalar distributive law:** for all $\lambda, \mu \in \mathbb{F}$ and $\mathbf{v} \in V$,

$$(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}.$$

10. **Vector distributive law** for all $\lambda \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$,

$$\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}.$$

Note: addition in the field and addition in the vector space is different, however this is not distinguished in notation.

- Axioms 1-5 say that $(V, +)$ is an abelian group
- Axioms 6-8 describe how $\mathbb{F}$ acts on V
- Axioms 9 and 10 describe that addition and scalar multiplication are compatible.

Note: each axiom (except for 4) are "for all" statements, and apply to all

1.3 Basic Properties

In any vector space V , the following properties hold

1. **Uniqueness of Zero:** There is one and only one zero vector

Proof: If $\mathbf{0}' \in V$ satisfies $\mathbf{v} + \mathbf{0}' = \mathbf{v}$ for all $\mathbf{v} \in V$, then

$$\mathbf{0}' = \mathbf{0}' + \mathbf{0} = \mathbf{0} + \mathbf{0}' = \mathbf{0}.$$

2. **Cancellation Property:** If $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ satisfy

$$\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}, \text{ then } \mathbf{v} = \mathbf{w}.$$

3. **Uniqueness of Negatives**

1.3.1 Additional basic properties

Suppose that $\mathbf{v}$ is a vector in a vector space V ; λ is a scalar; 0 is the zero scalar; $\mathbf{0}$ is the zero vector in V . Then the following properties hold.

- $\lambda\mathbf{0} = \mathbf{0}$,
- $0\mathbf{v} = \mathbf{0}$.
- $(-1)\mathbf{v} = -\mathbf{v}$. (Here, -1 is a scalar and $-\mathbf{v}$ is the additive inverse of $\mathbf{v}$).
- If $\lambda\mathbf{v} = \mathbf{0}$, then either $\lambda = 0$ or $\mathbf{v} = \mathbf{0}$.

- If $\lambda \mathbf{v} = \mu \mathbf{v}$, and $\mathbf{v} \neq \mathbf{0}$, then $\lambda = \mu$.

1.4 Standard Examples

Example 1

The set $\mathbb{F}^n = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} : a_i \in \mathbb{F} \forall i \right\}$ is a vector space over $\mathbb{F}$ under the usual (component-wise) addition and multiplication by a scalar.

Example 2

The set $M_{m,n}(\mathbb{F})$ of all $m \times n$ matrices with entries in $\mathbb{F}$ is a vector space over $\mathbb{F}$.

Example 3

Let X be any non-empty set. The set $\mathcal{F}[X] = \{f : X \rightarrow \mathbb{F}\}$ of all $\mathbb{F}$ -valued functions with domain X is a vector space over $\mathbb{F}$ with point-wise addition and scalar multiplication.

To prove that a system V is a vector space, every axiom must be verified. To prove that V is not a vector space, only one axiom needs to be disproven.

Chapter 2

Standard Examples and Subspaces

There are subsets of $\mathcal{R}[\mathbb{R}]$ that are also vector spaces.

The substructures of a field are called subfields. More precisely:

Definition

If $\mathbb{K}$ is a field, then a subset $\mathbb{F} \subseteq \mathbb{K}$ is called a subfield of $\mathbb{K}$, denoted by $\mathbb{F} \leq \mathbb{K}$, if $\mathbb{F}$ is also a field with the same $+$ and $\cdot$ as in $\mathbb{K}$.

Definition

Let V be a vector space over a field $\mathbb{F}$. A subset $S \subseteq V$ is called a subspace of V if S is itself a vector space over $\mathbb{F}$ with the same operations for addition and scalar multiplication.

If S is a subspace of V , then S contains the zero vector $\mathbf{0}_V$ of V . Hence, if $\mathbf{0}_V \notin S$, then S is not a subspace. I.e. a subspace must have the unique zero vector of its larger space.

Example

Consider $\mathbb{R}^3$ as a vector space over $\mathbb{R}$.

1. $\mathbf{0}$ is a subspace. $\mathbb{R}^3$ itself is a subspace.
2. Any line through the origin $\mathbf{0}$ is a subspace.
3. Any plane through the origin is a subspace.

These are the only subspaces in $\mathbb{R}^3$

2.1 The Subspace Theorem

Theorem

A subset S of a vector space V over $\mathbb{F}$ is a subspace if and only if

- $\mathbf{0} \in S$,
- $\mathbf{u} + \mathbf{v} \in S$ for all $\mathbf{u}, \mathbf{v} \in S$, and
- $\lambda \mathbf{v} \in S$ for all $\mathbf{v} \in S$, scalars $\lambda \in \mathbb{F}$.

Alternatively,

- $\mathbf{0}_V \in S$, and
- $\lambda \mathbf{u} + \mathbf{v} \in S$, and scalars $\lambda \in \mathbb{F}$.

Note: the only subspaces of $\mathbb{R}^2$ are lines that pass through the origin, ie. contain the zero vector.

Example

For which values $a \in \mathbb{R}$ is the set $S_a = \left\{ y \in \mathcal{R}[\mathbb{R}] : \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 4y = a \right\}$ a subspace of $\mathcal{R}[\mathbb{R}]$?

Since the zero vector is not defined for values of a other than 0, the only solution is $a = 0$.

Finding all subspaces of $\mathbb{F}^n$

Suppose $W \leq \mathbb{F}^n$. Then $\mathbf{0} \in W$. So

- either $W = \mathbf{0}$ or $\exists \mathbf{u} \in W$, $\mathbf{u} \neq \mathbf{0}$. Then all scalar multiples of $\mathbf{u}$ are in W . Thus, $\lambda \mathbf{u} \mid \lambda \in \mathbb{F}$
- Thus, either $W = \{\lambda \mathbf{u} \mid \lambda \in \mathbb{F}\}$ or $\exists \mathbf{v} \in W$ with $\mathbf{v} \neq \lambda \mathbf{u}$ for all $\lambda \in \mathbb{F}$. So $\{\lambda \mathbf{u} + \mu \mathbf{v} \mid \lambda, \mu \in \mathbb{F}\} \leq W$. Hence,
- either $W = \{\lambda \mathbf{u} + \mu \mathbf{v} \mid \lambda, \mu\}$ or $\exists \mathbf{w} \in W$ with $\mathbf{w} \neq \lambda \mathbf{u} + \mu \mathbf{v}$ for all λ, μ . Thus, $\{\lambda \mathbf{u} + \mu \mathbf{v} + \nu \mathbf{w} \mid \lambda, \mu, \nu \in \mathbb{F}\} \leq W$
- If $n = 3$, then $W = \{\lambda \mathbf{u} + \mu \mathbf{v} + \nu \mathbf{w} \mid \lambda, \mu, \nu \in \mathbb{F}\}$. If $n > 3$, we may continue this construction.

Chapter 3

Lin. Combination and Lin. Span

3.1 Linear Combination

Let V be a vector space. We can form a vector in V by adding up scalar multiples of finitely many vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in V . Such a vector is called a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$

Definition

A linear combination of finitely many vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ in a vector space V over $\mathbb{F}$ is a sum of scalar multiples of the form.

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \text{ with } \lambda_1, \dots, \lambda_n \in \mathbb{F}.$$

Example

Consider the subspace

$$S = \{p \in \mathbb{P}_2(\mathbb{R}) \mid xp'(x) = 2p(x) \forall x \in \mathbb{R}\} \leq \mathbb{P}_2(\mathbb{R})$$

3.2 Linear Span and Spanning Set

Definition

Let S be a subset of a vector space V . Then the span of S is the set $\text{span}(S)$ of all linear combinations of vectors in S :

$$\text{span}(S) = \left\{ \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \mid \begin{array}{l} \mathbf{v}_1, \dots, \mathbf{v}_n \in S \\ \lambda_1, \dots, \lambda_n \in \mathbb{F} \end{array}, \forall n \in \mathbb{N} \right\}$$

If $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, then we write $\text{span}(S) = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. If $\text{span}(S) = V$, S is called a spanning set of V .

Example

Let S be a nonempty subset of a vector space V . Then $\text{span}(S)$ is a subspace of V , and it is the smallest subspace containing S .

First, for any $\mathbf{v} \in S$, $\mathbf{0} = 0\mathbf{v} \in \text{span}(S)$. If $\mathbf{v}, \mathbf{u} \in \text{span}(S)$, then $\mathbf{v} = \lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n$ and $\mathbf{u} = \mu_1\mathbf{u}_1 + \cdots + \mu_m\mathbf{u}_m$ for some $\mathbf{v}_i, \mathbf{u}_i \in S$ and $\lambda_i, \mu_i \in \mathbb{F}$. Thus, if $\lambda \in \mathbb{F}$, then

$$\begin{aligned}\lambda\mathbf{v} + \mathbf{u} &= \lambda(\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n) + \mu_1\mathbf{u}_1 + \cdots + \mu_m\mathbf{u}_m \\ &= (\lambda\lambda_1)\mathbf{v}_1 + \cdots + (\lambda\lambda_n)\mathbf{v}_n + \mu_1\mathbf{u}_1 + \cdots + \mu_m\mathbf{u}_m\end{aligned}$$

is a linear combination of vectors in S . Hence, $\lambda\mathbf{v} + \mathbf{u} \in \text{span}(S)$, proving $\text{span}(S) \leq V$ by the alternative subspace test.

Example

What is the geometric interpretation of $\text{span} \left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$ in $\mathbb{R}^3$?

It is a plane through the origin with parametric vector form

$$\mathbf{x} = \lambda \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

Example

Let $\mathbf{u} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} \in \mathbb{R}^3$, and the set $S = \left\{ \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right\}$. Is $\mathbf{u} \in \text{span}(S)$

Check there exists scalars $\lambda, \mu \in \mathbb{R}$ such that

$$\lambda \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 1 \\ 3 & 2 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} \lambda \\ \mu \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

Thus, we need to test the solvability of a linear system $A\mathbf{x} = \mathbf{b}$ with augmented matrix $\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 3 & 2 & 1 \\ 4 & 3 & 0 \end{array} \right)$.

The matrix can be reduced into row echelon form

$$\left(\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & 0 \end{array} \right).$$

A solution to the system exists, meaning that $\mathbf{u} \in \text{span}(S)$. We substitute back to find λ and μ , providing $\mu = 3$ and $\lambda = 1 - \mu = -2$.

It can also be applied to find an arbitrary vectors. By solving the linear combination of the vectors in the span for $\mathbf{b}$, a generalised solution for the span can be found.

Suppose $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and a vector $\mathbf{b}$ are all in $\mathbb{F}^m$. Then $\mathbf{b} \in \text{span}(S)$ if and only if there are scalars $\mathbf{x}_1, \dots, \mathbf{x}_n$ such that

$$x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n = \mathbf{b}.$$

Chapter 4

Linear Dependence/Independence

Example

Does $\sin^2 x$ lie in the span of $\{1, \cos^2 x\}$ inside $C(\mathbb{R})$?

Yes, $\sin^2 x = 1 - \cos^2 x$

Example

Does e^x lie in the span of $\{1, x, x^2, \dots\}$ inside $C(\mathbb{R})$?

No, exponential functions grow faster than any polynomial.

Definition

The subspace of $\mathbb{F}^m$ spanned by the columns of an $m \times n$ matrix A is called the column space of A and is denoted by $\mathcal{C}(A)$. The subspace $\mathcal{N}(A)$ of the solutions of $A\mathbf{x} = \mathbf{0}$ is called the null space of A .

If a vector can be written as a scalar multiple of any of the columns in the matrix, it is in the column space.

Suppose $S \subset V$. If we can remove a vector from the set S without changing the span, then the set S is said to be linearly dependent. Otherwise the set is linearly independent.

If $\mathbf{0} \in S$, then S is linearly dependent.

If S is a spanning set of V and $\mathbf{v} \in V - S$, then $S \cup \{\mathbf{v}\} \dots$

Testing Linear Dependence/Independence

Example

$$S_1 = \left\{ \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}, \quad S_2 = \left\{ \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right\}, \quad S_3 = S_u \cup \left\{ \begin{pmatrix} 1 \\ -1 \\ 7 \end{pmatrix} \right\}$$

- S_1 is linearly dependent
- S_2 is linearly independent
- S_3 no vector is a scalar multiple of the other, but the sum of two make the third, therefore it is linearly dependent

Definition

Let S be a subset of a vector space.

1. If there exist distinct vectors $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$ and scalars $\lambda_1, \dots, \lambda_n$, not all zero, such that

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0},$$

then we say that S is linearly dependent.

2. If the opposite case is true, that is that the only solution for the linear combination of vectors in $S = \mathbf{0}$ is where the scalar multiple for every vector is 0, S is linearly independent.

Example

Recall the dot product for $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$:

$$\mathbf{u} \cdot \mathbf{v} := u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

In any inner product space, orthogonal vectors are linearly independent. In particular, if $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subset \mathbb{R}^n$ satisfies $\mathbf{v}_i \neq \mathbf{0}$ and $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for all $i \neq j$, then S is linearly independent.

Suppose

$$\sum_{i=1}^m x_i \mathbf{v}_i = \mathbf{0} \quad \text{for some } x_i \in \mathbb{R}$$

Then for any $1 \leq j \leq m$

$$0 = \mathbf{v}_j \cdot \mathbf{0} = \mathbf{v}_j \cdot \sum_{i=1}^m x_i \mathbf{v}_i = \sum_{i=1}^m x_i (\mathbf{v}_j \cdot \mathbf{v}_i) \stackrel{\perp}{=} x_j (\mathbf{v}_j \cdot \mathbf{v}_j)$$

As $\mathbf{v}_j \cdot \mathbf{v}_j \neq 0$, this proves that $x_j = 0$.

Definition

The third definition of linear dependence/independence is as follows:

Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$. Then

1. S is linearly independent if and only if the following condition holds: For all $\mathbf{v}_1, \dots, \mathbf{v}_n \in S$ and $\lambda_i, \mu_i \in \mathbb{F}$,

$$\text{if } \sum_{i=1}^n \lambda_i \mathbf{v}_i = \sum_{i=1}^n \mu_i \mathbf{v}_i, \text{ then } \lambda_i = \mu_i, \forall i.$$

2. S is linearly dependent if and only if $\exists \mathbf{u}_1, \dots, \mathbf{u}_n \in S$ and $\alpha_i, \beta_i \in \mathbb{F}$ such that $\sum_{i=1}^n \alpha_i \mathbf{u}_i = \sum_{i=1}^n \beta_i \mathbf{u}_i$ and $\alpha_i \neq \beta_i$ for at least one i .

4.1 Bases

Definition

A set $\mathcal{B}$ of vectors in a vector space V is called a basis for V if $\mathcal{B}$ is linearly independent and $V = \text{span}(\mathcal{B})$

- A minimal spanning set is a basis
- A maximal linearly independent set is a basis

A subset $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ of V forms a basis for V if and only if $\text{span}(\mathcal{B}) = V$ and every $\mathbf{v} \in V$ is uniquely expressible as

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n.$$

In $\mathbb{F}^n$, let $\mathbf{e}_i$ be the vector with the i^{th} entry 1 and all other entries 0. The set $\mathcal{S} = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is a linearly independent spanning set of $\mathbb{F}^n$. Hence, it is a basis; we call it the standard basis of $\mathbb{F}^n$. For example:

- In $\mathbb{F}^2$, the standard basis is $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$, and
- in $\mathbb{F}^3$, the standard basis is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$.

In $M_{m,n}(\mathbb{F})$, let E_{ij} be the matrix with the (i, j) -entry 1 and all other entries 0. The set $\mathcal{S} = \{E_{ij} | 1 \leq i \leq m, 1 \leq j \leq n\}$ is a linearly independent spanning set of $M_{m,n}(\mathbb{F})$. Hence, it is a basis which is called the standard basis of $M_{m,n}(\mathbb{F})$.

The set $\mathcal{S}_n = \{1, x, \dots, x^n\}$ is the standard basis for $\mathbb{P}_n(\mathbb{R})$